

ISHA PATRO

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EDUCATION

NEW YORK UNIVERSITY

Master of Science in **Financial Engineering**

GPA: 3.6/4.0

NY, NY

Expected 05/27

VELLORE INSTITUTE OF TECHNOLOGY, AP

Bachelor in **Computer Science**

GPA: 3.5/4.0

Andhra Pradesh, India

07/21

PROGRAMMING & TECHNICAL SKILLS

Programming & Development: Python, JavaScript, HTML, CSS, VBA, React, REST APIs, Streamlit, Dash, AG Grid, Tableau

Data Science: Statistical Modeling, Machine Learning, Deep Learning, Attention Mechanisms, Large Language Models (LLMs), Generative AI, RAG, Agentic AI, Model Context Protocol (MCP), Natural Language Processing (NLP), Fine-Tuning, AI Inference, LLM Evaluation, Prompt Engineering

Databases & Cloud: SQL, q/kdb+, AWS, MongoDB, Microsoft Office, Microsoft Excel, Bloomberg Terminal

Quantitative Finance: Fixed Income, Equity Research, Portfolio Management, Risk Management, Time Series Forecasting, Prediction Markets

EXPERIENCE

FINANCIAL ARTIFICIAL INTELLIGENCE WORKING GROUP

07/26 - 09/26

Quantitative Finance Research Intern

- Developed a corpus for Fixed Income Modelling, Credit Card Securitization, and Derivatives workflows to benchmark AI agent outputs against Excel-based models, improving calculation accuracy, reasoning traceability, and hallucination resistance.

J.P. MORGAN CHASE & CO. | Commercial & Investment Bank

01/21 - 08/25

Software Developer Engineer II, MBS Division - Fixed Income Pricing Team (Bangalore, India)

- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting \$5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

PUBLICATIONS / OPEN SOURCE CONTRIBUTION

Patro, Isha (2026) "Modeling Volatility in Polymarket Prices with Binary Jump-Tweedie Dynamics," Under review for the Prediction Markets Conference, Yale School of Management & Swensen Asset Management Institute.

Patro, Isha and Appana, Raj (2025) "FinOKF: Evidence-Aware Finance Open Knowledge Formatting with LRU Caching for Auditable Equity Research," Under review at Journal of Association for the Advancement of Artificial Intelligence.

Patro, Isha (2026) "Implied Prepayment Surfaces via Finance-Informed Neural Networks in Agency MBS Markets," Under review at Journal of Computational Finance.

Patro, Isha (2024) "[Unlocking Financial Wisdom: Building Knowledge Graphs from Investopedia](#)," *International Journal of Research and Analytical Reviews (IJRAR)*, Sep 2024.

Patro, Isha (2025) "[qutePandas: A Pandas-Compatible Execution Layer Backed by kdb+/q](#), Delivering 10x–100x+ Speedups on Large-Scale Analytical Workloads," Open Source Contribution. Published on PyPI, June 2025.

HONORS / AWARDS

- Awarded the prestigious **Numerix Women in Finance Scholarship**, a **\$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (**\$1,200**) - across all engineering disciplines, 2017-2018

ACADEMIC PROJECTS

Spline-Based Conditional Market Beta & Statistical Arbitrage Framework

09/26

- Modelled a NumPy-based cubic spline regression and k-fold cross-validated truncated basis engine to capture state-dependent non-linear asset co-movements, extracting idiosyncratic residual spreads for market-neutral statistical arbitrage.

Nike Equity Research Dashboard | DCF Valuation & Competitive Channel Analysis

09/26

- Built a data-driven equity research platform integrating SEC filings, DCF valuation, peer benchmarking, sentiment-adjusted price targets, and wholesale-channel analysis across Deckers, On, and Foot Locker.

Money Talks

09/26

- Engineered an AI-powered SaaS analytics platform that surfaces revenue concentration risk, tracks high-value customer churn, and evaluates go-to-market performance alongside underlying monthly recurring revenue (MRR) health.

2WIN

06/26

- Designed and backtested quantitative prediction-market strategies using tick-level order-book data, dynamic risk-based sizing, automated execution logic, and portfolio PnL attribution across real-time Polymarket markets.

VORTEX

05/26

- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.

Deep Temporal Models With Attention and Skip Connections

03/26

- Architected an end-to-end commodity intelligence pipeline combining FRED macroeconomic data, FinBERT news embeddings, dynamic graph learning, hybrid hyperbolic attention, and deep temporal models for futures-direction forecasting.

Geometry of Intelligence

01/25 - Present

- Created an independent machine learning research series explaining algorithms from first principles through mathematical intuition, probability, and clean code implementations.

Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation

01/26 - 03/26

- Analyzed cross-currency basis and liquidity fragmentation in crypto markets using high-frequency data across exchanges, evaluating stablecoin-driven pricing inefficiencies and regulatory impact (GENIUS Act)

Monte Carlo Pricing of European Barrier Options

10/25

- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.

Fixed Income Analytics & Securitization Suite

09/25

- Built a financial modeling suite in Excel featuring Yield Curve Calibration using Nelson-Siegel model, Mortgage-Backed Securities (MBS) delta-neutral hedging, and multi-tranche credit card ABS cash flow waterfalls.

Advanced NASDAQ Market Prediction System with Deep Learning Optimization

08/25

- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.

Minted Wealth

03/25

- Developed a comprehensive financial intelligence dashboard integrating live market data, portfolio performance analytics, and net worth insights for smarter investment decisions.